

WaRee11 v1.55

Technical Reference Manual

River Simulation and Hydraulic Modeling for iPad

Draft Reference Document

Prepared for WaRee11 v1.55 Review and Documentation

June 4, 2026

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Preface

This document is a draft technical reference for WaRee11 v1.55. It is intended to support users, reviewers, testers, collaborators, and future developers who need to understand the modeling concepts, hydraulic assumptions, data structure, numerical workflow, diagnostics, and result interpretation used in the application.

Unlike the user manual, which focuses on step-by-step operation, this reference manual focuses on the technical background and internal logic behind the application. It may be used together with the user manual, white paper, TestFlight review notes, and future academic or engineering documentation.

Reference Note

This reference is a draft. Final equations, solver options, screen names, and data structures should be checked against the final WaRee11 v1.55 source code before publication.

Revision History

Revision	Date	Description
0.1	Draft	Initial WaRee11 v1.55 technical reference structure.

1 Scope of the Reference Manual

1.1 Purpose

The purpose of this reference manual is to describe the technical foundation of WaRee11 v1.55. It explains how the application organizes a river modeling project, how data are represented, what hydraulic concepts are used, and how the simulation workflow is prepared for model readiness and result review.

1.2 Relationship to the User Manual

The user manual explains how to use the application. This reference manual explains why each model component is needed and how it contributes to the hydraulic simulation workflow.

Document	Main Purpose
User Manual	Step-by-step user operation and screen guidance.
Reference Manual	Technical explanation of model concepts, equations, assumptions, diagnostics, and internal workflow.
White Paper	Product vision, research motivation, and broader technical positioning.
Developer Notes	Source-code implementation details and future engineering tasks.

1.3 Intended Readers

This reference is intended for:

- Hydraulic engineering students.
- River modeling instructors.
- Research collaborators.
- TestFlight reviewers and technical evaluators.
- Future WaRee11 developers.
- Engineers comparing WaRee11 with desktop modeling tools.

2 Application Concept

2.1 Overview

WaRee11 is an iPad-based application for one-dimensional river hydraulic modeling. It is designed to simplify the process of preparing, checking, running, and reviewing river model simulations on a mobile device.

The application is not intended to replace mature desktop hydraulic packages in its early versions. Instead, it provides a mobile-first modeling environment for education, demonstration, field data preparation, conceptual analysis, and early-stage hydraulic model setup.

2.2 Design Philosophy

The design philosophy of WaRee11 v1.55 is based on the following principles:

- **Summary-first workflow:** show the most important information first.
- **Model readiness:** guide users toward complete and valid model setup.
- **Transparent diagnostics:** explain why a simulation can or cannot run.
- **Educational clarity:** expose key hydraulic concepts in a readable way.
- **iPad-native interaction:** use touch-friendly navigation and clear screen hierarchy.

2.3 Core Modeling Workflow

The modeling workflow is:

1. Create or open a project.
2. Define project metadata.
3. Build the river network.
4. Define cross sections.
5. Assign roughness and hydraulic properties.
6. Define boundary conditions.
7. Configure simulation settings.
8. Run readiness diagnostics.
9. Execute simulation.
10. Review results and export outputs.

3 Hydraulic Modeling Background

3.1 One-Dimensional River Modeling

A one-dimensional river model represents flow along the main direction of a river or channel. Hydraulic variables such as discharge and water level are computed along a river centerline, usually at cross-section locations or computational cells.

The one-dimensional assumption is suitable when the main interest is longitudinal variation of flow and water level. It is commonly used for river routing, flood level estimation, educational modeling, and conceptual hydraulic analysis.

3.2 Key Hydraulic Variables

Variable	Symbol	Description
Discharge	Q	Flow rate through a cross section.
Water level	H	Elevation of the water surface.
Flow depth	h	Water depth above bed elevation.
Flow area	A	Wetted flow area.
Wetted perimeter	P	Length of wetted boundary.
Hydraulic radius	R	Ratio of area to wetted perimeter.
Velocity	u	Mean flow velocity.
Manning roughness	n	Empirical channel resistance coefficient.
Bed slope	S_0	Longitudinal bed slope.
Friction slope	S_f	Energy slope due to resistance.

3.3 Continuity Principle

The continuity principle expresses conservation of mass. In a simplified one-dimensional form:

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = q_l \quad (3.1)$$

where:

- A is flow area,
- Q is discharge,
- x is distance along the river,
- t is time,
- q_l is lateral inflow per unit length.

3.4 Momentum Principle

The one-dimensional momentum equation expresses conservation of momentum. A common Saint-Venant form is:

$$\frac{\partial Q}{\partial t} + \frac{\partial}{\partial x} \left(\frac{Q^2}{A} \right) + gA \frac{\partial H}{\partial x} = gA(S_0 - S_f) \quad (3.2)$$

where g is gravitational acceleration.

Reference Note

Different hydraulic models use different formulations and simplifications of the Saint-Venant equations. WaRee11 should document the exact formulation used by each implemented solver.

4 Cross-Section Geometry Reference

4.1 Role of Cross Sections

Cross sections define the channel shape at selected river locations. They are essential because hydraulic area, conveyance, water depth, and storage depend on the geometry of each section.

4.2 Station-Elevation Data

A cross section is usually represented by a sequence of station-elevation points:

$$(x_i, z_i), \quad i = 1, 2, \dots, N \quad (4.1)$$

where x_i is the horizontal station and z_i is the bed elevation.

4.3 Area Calculation

For a given water level H , submerged portions of the cross section contribute to the wetted flow area. In a simplified trapezoidal segment between two points, the local water depths are:

$$d_i = \max(0, H - z_i) \quad (4.2)$$

The approximate area between two adjacent stations may be computed as:

$$A_i = \frac{d_i + d_{i+1}}{2} (x_{i+1} - x_i) \quad (4.3)$$

The total flow area is:

$$A = \sum_i A_i \quad (4.4)$$

4.4 Wetted Perimeter

The wetted perimeter is the length of the channel boundary in contact with water. For each wet segment:

$$P_i = \sqrt{(x_{i+1} - x_i)^2 + (z_{i+1} - z_i)^2} \quad (4.5)$$

The total wetted perimeter is:

$$P = \sum_i P_i \quad (4.6)$$

4.5 Hydraulic Radius

The hydraulic radius is:

$$R = \frac{A}{P} \quad (4.7)$$

4.6 Conveyance and Rating Curve

Using Manning's equation, discharge can be estimated as:

$$Q = \frac{1}{n} A R^{2/3} S^{1/2} \quad (4.8)$$

This relation can be used to generate a rating curve showing discharge as a function of water level.

Implementation Note

For implementation, the application should guard against invalid geometry, zero wetted perimeter, negative depth, and insufficient station-elevation points.

5 River Network Reference

5.1 River Network Representation

A river network defines how river branches, nodes, and junctions are connected. A simple river model may contain one branch, while a more complex model may include tributaries and confluences.

5.2 Main Objects

Object	Reference Description
Project	Container for all model data and results.
River Branch	A river reach from upstream to downstream.
Node	A location in the network, such as branch endpoint or junction.
Junction	A node where two or more branches connect.
Cross Section	Geometry assigned to a position along a branch.
Boundary Condition	External hydraulic condition at model boundary.
Simulation Setting	Numerical and runtime configuration.
Result Record	Output variable generated by a simulation.

5.3 Flow Direction

Each branch should have a defined flow direction from upstream to downstream. Chainage should be consistent with the selected orientation.

5.4 Network Validation Rules

Recommended validation rules include:

- At least one river branch exists.
- Each branch has upstream and downstream nodes.
- Cross sections are assigned to valid branches.
- Chainage values are ordered or otherwise clearly defined.
- Boundary conditions are assigned to model boundaries.
- Junctions are connected to valid branches.

6 Boundary Condition Reference

6.1 Role of Boundary Conditions

Boundary conditions define the hydraulic behavior at the edges of the computational domain. A typical one-dimensional river model requires at least one upstream and one downstream boundary condition.

6.2 Boundary Condition Categories

Category	Description
Constant Discharge	Fixed inflow or outflow discharge.
Constant Water Level	Fixed upstream or downstream stage.
Time-Series Discharge	Hydrograph-defined discharge varying with time.
Time-Series Water Level	Stage hydrograph varying with time.
Rating Curve Boundary	Boundary derived from discharge-stage relation.
Normal Depth Boundary	Downstream condition estimated from slope and roughness.

6.3 Time-Series Interpolation

For a time-series boundary condition, values may be interpolated between known data points. Linear interpolation between (t_1, y_1) and (t_2, y_2) is:

$$y(t) = y_1 + \frac{y_2 - y_1}{t_2 - t_1}(t - t_1) \quad (6.1)$$

6.4 Coverage Check

The time-series data should cover the full simulation period:

$$t_{\min} \leq t_{\text{start}} \quad \text{and} \quad t_{\max} \geq t_{\text{end}} \quad (6.2)$$

Important

If boundary data do not cover the simulation period, the model may stop, extrapolate, or produce unreliable results. WaRee11 should clearly report this condition before running.

7 Numerical Solver Reference

7.1 Solver Purpose

The solver computes hydraulic variables through time and space. In WaRee11, the solver is responsible for updating water level, depth, area, velocity, and discharge based on the selected numerical method and model configuration.

7.2 State Variables

A simplified finite-volume state vector may be written as:

$$\mathbf{U} = \begin{bmatrix} A \\ Q \end{bmatrix} \quad (7.1)$$

where A is area and Q is discharge.

7.3 Conservative Form

A simplified conservative formulation can be expressed as:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} = \mathbf{S}(\mathbf{U}) \quad (7.2)$$

where $\mathbf{F}$ is the flux vector and $\mathbf{S}$ is the source term vector.

7.4 Finite-Volume Update

A general finite-volume update for cell i is:

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^n - \frac{\Delta t}{\Delta x} (\mathbf{F}_{i+1/2} - \mathbf{F}_{i-1/2}) + \Delta t \mathbf{S}_i \quad (7.3)$$

7.5 Numerical Flux

The numerical flux estimates flow between neighboring cells. Examples include:

- Rusanov flux.
- HLL flux.
- HLLC flux.
- MacCormack predictor-corrector scheme.
- Lax-Wendroff scheme.

7.6 Rusanov Flux

A common Rusanov flux form is:

$$\mathbf{F}_{i+1/2} = \frac{1}{2} [\mathbf{F}(\mathbf{U}_L) + \mathbf{F}(\mathbf{U}_R)] - \frac{1}{2} a_{\max} (\mathbf{U}_R - \mathbf{U}_L) \quad (7.4)$$

where $a_{\max}$ is the maximum local wave speed.

7.7 HLL Flux

The Harten-Lax-van Leer flux uses estimated left and right wave speeds S_L and S_R :

$$\mathbf{F}_{HLL} = \frac{S_R \mathbf{F}_L - S_L \mathbf{F}_R + S_L S_R (\mathbf{U}_R - \mathbf{U}_L)}{S_R - S_L} \quad (7.5)$$

for $S_L < 0 < S_R$.

7.8 Source Terms

Source terms may include:

- Bed slope.
- Friction slope.
- Lateral inflow.
- External control terms.

7.9 Friction Slope

Using Manning's formula, a common friction slope expression is:

$$S_f = \frac{n^2 Q |Q|}{A^2 R^{4/3}} \quad (7.6)$$

Implementation Note

The solver should protect against division by zero, very small depth, negative area, and nonphysical discharge values.

8 Stability and Time-Step Reference

8.1 Courant-Friedrichs-Lewy Condition

For explicit solvers, the time step must be selected carefully. A common stability condition is based on the Courant number:

$$C = \frac{a\Delta t}{\Delta x} \quad (8.1)$$

where a is a representative wave speed.

8.2 Wave Speed

For shallow-water type equations, wave speed is often approximated by:

$$a = |u| + \sqrt{gh} \quad (8.2)$$

where:

- u is velocity,
- g is gravitational acceleration,
- h is flow depth.

8.3 Recommended Time Step

A recommended time step may be estimated as:

$$\Delta t \leq C_{target} \frac{\Delta x}{|u| + \sqrt{gh}} \quad (8.3)$$

8.4 Stability Diagnostics

The application should report:

- Estimated maximum velocity.
- Estimated maximum wave speed.
- Recommended time step.
- User-selected time step.
- CFL status.

9 Data Model Reference

9.1 Project Data

A project is the top-level data container. It may include:

- Project metadata.
- River network.
- Cross-section collection.
- Boundary conditions.
- Simulation settings.
- Diagnostics.
- Results.
- Scenario snapshots.

9.2 Suggested Data Schema

The following schema is a conceptual reference and should be aligned with the actual Swift implementation.

Listing 9.1: Conceptual project model

```
struct WaReeProject {  
    var id: UUID  
    var name: String  
    var riverName: String  
    var studyArea: String  
    var createdAt: Date  
    var modifiedAt: Date  
    var network: RiverNetwork  
    var crossSections: [CrossSectionData]  
    var boundaryConditions: [BoundaryCondition]  
    var simulationSettings: SimulationSettings  
    var latestResults: [SimulationResult]  
}
```

9.3 Cross Section Model

Listing 9.2: Conceptual cross-section model

```
struct CrossSectionData {  
    var id: UUID  
    var name: String  
    var branchId: UUID  
    var chainage: Double  
    var points: [CrossSectionPoint]  
    var manningN: Double  
    var initialWaterLevel: Double?  
}
```

9.4 Boundary Condition Model

Listing 9.3: Conceptual boundary condition model

```
struct BoundaryCondition {  
    var id: UUID  
    var location: BoundaryLocation  
    var type: BoundaryConditionType  
    var constantValue: Double?  
    var timeSeries: [TimeSeriesPoint]  
}
```

9.5 Simulation Settings Model

Listing 9.4: Conceptual simulation settings model

```
struct SimulationSettings {  
    var timeStep: Double  
    var duration: Double  
    var outputInterval: Double  
    var scheme: NumericalScheme  
    var courantTarget: Double  
}
```

10 Diagnostics Reference

10.1 Purpose

Diagnostics help the user understand whether the model is ready to run. They also provide teaching value by explaining why each model component is required.

10.2 Diagnostic Severity Levels

Severity	Meaning
Info	General information or recommendation.
Warning	The model may run, but the user should review the issue.
Error	The model should not run until the issue is fixed.
Success	The item is complete or valid.

10.3 Readiness Categories

Recommended diagnostic categories include:

- Project metadata.
- River network.
- Cross-section geometry.
- Roughness values.
- Boundary conditions.
- Time-series coverage.
- Simulation settings.
- CFL and stability.
- Result availability.

10.4 Example Diagnostic Messages

Category	Severity	Message
Network	Error	No river branch has been defined.

Cross Section	Warning	Cross section CS-02 has fewer than recommended geometry points.
Boundary Condition	Error	Downstream boundary condition is missing.
Stability	Warning	Selected time step may exceed recommended CFL limit.
Results	Info	No simulation result is available yet.

11 Results Reference

11.1 Result Variables

Simulation outputs may include:

Variable	Symbol	Unit
Water level	H	m
Discharge	Q	m ³ /s
Depth	h	m
Velocity	u	m/s
Area	A	m ²
Froude number	Fr	–

11.2 Froude Number

The Froude number is:

$$Fr = \frac{u}{\sqrt{gh}} \quad (11.1)$$

It indicates flow regime:

- $Fr < 1$: subcritical flow.
- $Fr = 1$: critical flow.
- $Fr > 1$: supercritical flow.

11.3 Result Interpretation

Users should check:

- Whether water levels are physically reasonable.
- Whether discharge changes smoothly unless a sudden boundary change is expected.
- Whether negative depth occurs.
- Whether oscillation appears in the result chart.
- Whether warnings were produced during simulation.

12 Scenario and Comparison Reference

12.1 Scenario Concept

A scenario is a saved model case. It allows users to compare different assumptions or settings.

Examples of scenario changes include:

- Different Manning roughness.
- Different upstream hydrograph.
- Different downstream water level.
- Different numerical scheme.
- Different simulation duration or time step.

12.2 Scenario Comparison

Scenario comparison can be used to answer questions such as:

- How sensitive is water level to Manning roughness?
- How does a flood hydrograph affect peak water level?
- How much does the downstream boundary influence the result?
- Does a smaller time step improve stability?

13 Export Reference

13.1 Exported Result Data

A result CSV file should contain enough information to identify the model output clearly:

Field	Description
Project ID	Unique project identifier.
Scenario ID	Unique scenario identifier, if available.
Time	Simulation time.
Location ID	Cross section or cell location.
Chainage	Distance along river branch.
Water Level	Computed water surface elevation.
Discharge	Computed flow rate.
Depth	Computed flow depth.
Velocity	Computed mean velocity.
Warning Flag	Optional diagnostic flag.

13.2 Diagnostic Export

Diagnostic export should include:

- Diagnostic category.
- Severity.
- Message.
- Suggested correction.
- Timestamp.

14 Reference Implementation Notes

14.1 Architecture Direction

The recommended architecture for WaRee11 is:

- SwiftUI for interface.
- MVVM for screen logic.
- Repository pattern for project data.
- Actor-based services for safe async data operations.
- Dedicated solver engine independent from UI.
- Export services separated from result views.

14.2 Recommended Module Boundaries

Module	Responsibility
Core Models	Project, network, cross section, boundary condition, settings, result types.
Repositories	Loading, saving, cloning, and deleting model data.
Solver Engine	Hydraulic calculation and numerical update.
Diagnostics	Readiness checks and validation messages.
Feature Views	User-facing SwiftUI screens.
Export Services	CSV, report, and share-sheet output.

14.3 Testing Recommendations

Recommended test groups include:

- Geometry calculation tests.
- Manning rating curve tests.
- Boundary interpolation tests.
- CFL advisor tests.
- Solver stability tests.
- Diagnostics completeness tests.
- Export format tests.
- UI workflow tests.

15 Limitations and Assumptions

15.1 Modeling Limitations

The current reference assumes WaRee11 is primarily a one-dimensional river modeling application. Therefore, the following limitations may apply:

- Lateral floodplain flow may not be fully represented.
- Complex two-dimensional flow patterns are outside the basic 1D workflow.
- Sediment transport and water quality may not be included.
- Calibration tools may be limited in early versions.
- Field data import/export may require additional development.

15.2 Software Limitations

Possible software limitations include:

- Early-stage solver support.
- Limited project import/export compatibility.
- Limited GIS editing functionality.
- Limited professional calibration and reporting functions.
- App Store/TestFlight feature restrictions depending on release stage.

15.3 Recommended Use

WaRee11 v1.55 is recommended for:

- Teaching hydraulic modeling concepts.
- Demonstrating 1D river modeling workflows.
- Testing mobile-first model setup.
- Early-stage conceptual river simulation.
- Field-friendly project preview and data review.

16 Future Development Reference

16.1 Potential Future Features

Future versions may include:

- More advanced finite-volume solvers.
- Calibration and optimization tools.
- Observed versus simulated comparison.
- GIS editing layer inside Network and Cross Section modules.
- Import from GeoJSON, CSV, and external hydraulic model formats.
- 2D or coupled 1D–2D workflows.
- Cloud project synchronization.
- Collaboration and reviewer mode.

16.2 GIS Editing Layer Direction

A future WaRee11 GIS editing layer may include:

- Map canvas.
- Layer manager.
- River centerline editing.
- Cross-section location editing.
- Attribute table editing.
- GeoJSON and CSV interoperability.
- Hydraulic model readiness checks.

A Symbol Reference

Symbol	Meaning	Unit
A	Flow area	m^2
Q	Discharge	m^3/s
H	Water level	m
h	Flow depth	m
u	Velocity	m/s
g	Gravitational acceleration	m/s^2
n	Manning roughness	$\text{s}/\text{m}^{1/3}$
R	Hydraulic radius	m
P	Wetted perimeter	m
S_0	Bed slope	–
S_f	Friction slope	–
Δt	Time step	s
Δx	Spatial step	m
C	Courant number	–
Fr	Froude number	–

B Suggested Figures for Final Reference

The final reference manual may include the following figures:

1. WaRee11 v1.55 system workflow diagram.
2. Project data structure diagram.
3. River network object diagram.
4. Cross-section geometry diagram.
5. Boundary condition time-series diagram.
6. Finite-volume cell diagram.
7. CFL stability explanation diagram.
8. Diagnostics and readiness workflow diagram.
9. Result export workflow diagram.

C Suggested References

The following references should be reviewed and formatted properly in the final version:

- LeVeque, R. J. *Finite Volume Methods for Hyperbolic Problems*. Cambridge University Press.
- MIKE 11 Reference Manual.
- D-Flow FM User Manual.
- SOBEK User Manual.
- HEC-RAS Hydraulic Reference Manual.
- Chow, V. T. *Open-Channel Hydraulics*.
- Cunge, J. A., Holly, F. M., and Verwey, A. *Practical Aspects of Computational River Hydraulics*.

Closing Note

This draft technical reference provides a foundation for documenting WaRee11 v1.55. It should be updated together with the final application source code, user interface, solver implementation, and App Store/TestFlight release notes.